

SPAM EMAIL CLASSIFICATION

USING NAIVE BAYES AND SUPPORT VECTOR MACHINE

Machine Learning project focused on detecting spam and ham messages using supervised text classification.

COURSE	MACHINE LEARNING
COURSE CODE	25SC2107E
BRANCH	CSE
SECTION	4

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PROJECT ABSTRACT

Spam classification is an important machine learning application for improving digital communication security and reducing unwanted, misleading and fraudulent messages. This project uses the spam.csv dataset containing labelled ham and spam text messages. The main objective is to build an automated classification framework that can distinguish legitimate messages from spam using natural language preprocessing, feature extraction and supervised machine learning.

The text is cleaned and transformed into numerical vectors using CountVectorizer and TF-IDF. Naive Bayes and Support Vector Machine are then trained and compared using Accuracy, Precision, Recall, F1-Score and Confusion Matrix. The expected outcome is an efficient and interpretable model that reduces false spam detection while improving reliable message filtering.

5,572
RAW MESSAGES

2
CLASSES

TF-IDF
FEATURES

NB/SV
M
MODELS

WHY SPAM DETECTION?

Unsolicited bulk messages affect productivity, storage management and cyber security. Spam messages often include advertisements, phishing attempts, fraudulent links and misleading content. Since the volume of digital messages is high, manual filtering is slow and unreliable. Therefore, an automated machine learning approach is useful for identifying suspicious text patterns and separating spam from legitimate messages.

ML WORKFLOW

The project follows a standard text-classification pipeline. The raw message is cleaned, converted into numerical vectors using CountVectorizer and TF-IDF, and then passed to Naive Bayes and Support Vector Machine classifiers. Comparing these models helps identify which method gives stronger spam detection with lower false positives and false negatives.



REQUIREMENTS	SPECIFICATIONS
Operating System & IDE	Windows 10/11 or Ubuntu Linux; Jupyter Notebook, Google Colab or Visual Studio Code
Programming & Libraries	Python 3.10+ , Scikit-learn, Pandas, NumPy, Matplotlib and Seaborn
Dataset & Storage	spam.csv dataset containing labelled ham/spam text messages; CSV-based storage
ML Pipeline	Text cleaning, CountVectorizer / TF-IDF Vectorizer, Naive Bayes and Support Vector Machine
Evaluation & Hardware	Accuracy, Precision, Recall, F1-Score and Confusion Matrix; 4 GB RAM minimum; GPU not required

PAPERS 1–5

Sno	Author Names	Title of the Paper	Year of Published	Dataset Used	Methodology Used	Limitation of the Paper
1	Nouha Arfaoui	Enhancing Machine Learning Algorithms for Imbalanced Data—A Case Study: Spam Detection	2026	SpamAssassin Public Corpus (Kaggle): easy_ham 2,551; hard_ham 250; spam_2 1,397. easy_ham + hard_ham are combined as ham; spam_2 is spam.	Text preprocessing with tokenization, stop-word removal and lemmatization; stratified 80/20 split; CountVectorizer and TF-IDF; six balancing methods (SMOTE, BorderlineSMOTE, RandomUnderSampler, TomekLinks, SMOTEENN, SMOTETomek); classifiers include MultinomialNB, LinearSVC, KNN, Decision Tree, Logistic Regression, Random Forest and Gradient Boosting; GridSearchCV and RepeatedStratifiedKFold used for robust evaluation.	No dedicated limitations section is provided. The experimental scope is limited to one SpamAssassin corpus and accuracy is the primary evaluation metric, so cross-corpus generalization is not demonstrated.
2	Worawit Kitikulsoun; Nawaporn Wisitpongphan	Adaptive Semi-Personalized Email Classification Model (ASPEC) with Incremental Learning	2026	Anonymized real corporate email-server logs from a large Thai financial-services enterprise: 625,069 emails across 3 months and 9,788 employees in 12 functional departments. 423,043 emails used for training/testing; 202,026 for incremental-learning evaluation.	ASPEC uses privacy-preserving email metadata rather than body content. Classical ML models (Logistic Regression, Decision Tree, Random Forest) are combined with adaptive feature selection and group-level model selection based on occupational and behavioral context. Models are periodically updated using batch-incremental learning to adapt to changing communication patterns.	No direct comparison with transformer/deep-learning models; body content is excluded because only metadata is used; current grouping may need extension for very large enterprises; evaluation is from a single organization and limited time period, so cross-organization generalizability remains unvalidated.
3	Zhiyong He; Chang Liu; Boyang Hu; Duo Jia	SpamLLM: Leveraging Large Language Models for Robust Spam Email Classification	2026	Four public spam benchmarks: SpamAssassin, LingSpam, SMS5spamCollection and CSDMC2010.	Multimodal SpamLLM framework combines structured email metadata, body statistics, raw body text and attachment metadata. Numerical features are normalized, categorical features frequency-encoded, and text is represented with TF-IDF plus deep semantic features from a frozen LLaMA-3.1-8B backbone. Fused representations are processed by a lightweight classification head; training uses Adam with the LLM frozen, plus efficiency techniques such as FP16 and precomputed.	Evaluated on four relatively small/static benchmarks; adversarial spam, multilingual spam and cross-domain generalization were not empirically validated; freezing the backbone limits domain adaptation; the 8B-parameter model still has substantial memory and inference-latency requirements.
4	Sahar Hamad Ahmed; Hadeel M. Saleh; Sameer Alani	Enhanced Email Spam Filtering Using Spectral Clustering and Deep Neural Networks	2026	Enron Email Dataset; SpamAssassin Dataset; UCI Mail Spam Dataset.	Proposes SCDNN, combining spectral clustering with deep neural networks. Spectral clustering partitions the email corpus into neighborhood/centroid-based groups, then a DNN performs spam classification using distance/likelihood information. The approach is compared with backpropagation neural networks, SVM, Random Forest, Bayesian methods and other standard classifiers using accuracy, precision, recall and F1-score.	The authors state that further investigation is required for complex deployment scenarios such as cloud computing and real-time systems, and that the method still needs optimization for big-data settings.
5	Dharmaraj R. Patil	Real-time Spam Email Filtering with Stacking and Majority Voting Ensembles	2026	Kaggle Email Spam Dataset.	NLP preprocessing removes HTML tags, special characters, URLs and stopwords and applies lemmatization. Text is converted using TF-IDF and CountVectorizer. Base classifiers are Logistic Regression, Decision Tree, SVM and Stochastic Gradient Descent; their predictions are combined using stacking and majority-voting ensembles. Evaluation uses accuracy, precision, recall, F-measure, FPR and FNR.	Performance depends strongly on training-data quality and diversity; stacking/voting add computational overhead; large-scale deployment introduces scalability challenges; static training data can become outdated as spam evolves; surface-level features can miss nuanced, obfuscated or semantically complex spam.

Literature papers are grouped exactly like the Excel sheet: five studies per slide, showing authors, title, year, dataset, methodology and limitation.

PAPERS 6–10

SNO	AUTHORS	TITLE	YEAR	DATASET	METHODOLOGY	LIMITATION
6	Abdulhanan Sharafat; Samet Aymaz; Cemal Köse	A Hybrid CNN-BiGRU-Attention Architecture for Enhanced Spam Email Classification	2026	TREC 2007 Public Spam Corpus (75,419); Enron-Spam (33,716); Spam-Assassin (5,796); Ling-Spam (2,893).	Email headers are removed and body text is normalized; URLs, email addresses and numbers are replaced with placeholders; text is lowercased and tokenized with a 10,000-word vocabulary, padded to 250 tokens, and represented with pretrained GloVe embeddings. The hybrid architecture combines CNN for local lexical/n-gram features, BiGRU for bidirectional long-range dependencies and an attention mechanism for salient tokens. Evaluation includes stratified holdout, five-fold cross.	Performance degrades under domain shift in cross-dataset testing; robust deployment would require more diverse multi-source corpora, domain adaptation or periodic retraining. Public labeled spam corpora are largely historical because of privacy constraints and may not reflect modern spam. Attention weights are not treated as post-hoc explanations.
7	Mahmoud E. Ali; Anjali Diwan; Rajendrasinh Jadeja	Binary Spam Detection Using GRPO and LoRA on Qwen 2.5-3B-Instruct	2026	SpamClassify (5,572); SpamDetection (10,900); LingSpam (2,893); SpamAssassin (10,749).	Fine-tunes Qwen 2.5-3B-Instruct for spam/ham classification using 4-bit quantization, LoRA and Group Relative Policy Optimization (GRPO). Datasets are standardized into instruction prompts with stratified train/test splits. Reward functions strongly reward correct classification and single-word outputs while penalizing wrong or verbose responses; GRPO uses grouped samples, reward scoring, relative advantage estimation and policy updates.	The study is limited to binary spam classification; multilingual, domain-specific, multiclass and multilabel settings remain untested. The reward design intentionally sacrifices explanatory responses in favor of one-word labels, and future work is needed across other model architectures and domains.
8	Shruti Raju Mahitkar	Explainable AI-Based Spam Email Classification System	2026 (PDF metadata; publication year not stated in article text)	Publicly available spam datasets plus user-submitted emails; the paper does not identify exact dataset names.	Email text is cleaned by removing HTML/noise, tokenization, stop-word removal and stemming or lemmatization. Bag of Words and TF-IDF are used for feature extraction; classifiers discussed include Naive Bayes, Logistic Regression and SVM. The proposed system emphasizes TF-IDF + Logistic Regression with SHAP-based word-level explanations and a dashboard for confidence scores and explanations; performance is evaluated with accuracy, precision, recall and F1-score.	Accuracy depends on training-data size, quality and diversity; false positives and false negatives remain possible; evolving/obfuscated/image-based spam can reduce effectiveness; frequent retraining increases maintenance cost; multilingual and mixed-language emails are challenging; explainability and privacy remain concerns.
9	Maram Bani Younes; Ahmad Ababneh	Machine Learning Based Spam Detection in Digital Communication Systems: A Comparative Analysis	2026	Nine benchmarks across three platforms: Enron (28,063), TREC (83,446), SpamAssassin (6,048), Ling-Spam (2,859), UCI SMS (5,169), Nigara SMS (5,171), Kaggle SMS (10,286), Twitter (5,169) and YouTube (350).	Uses public benchmark datasets for email, SMS and social media. Cleaning removes missing/duplicate records, normalizes text, removes punctuation/symbols and stopwords, and applies SMOTE to imbalanced training data. TF-IDF (top 500 features) is the primary representation. Six shallow ML classifiers—Naive Bayes, SVM, Random Forest, Decision Tree, KNN and Logistic Regression—are compared within-domain and across unseen datasets; additional experiments assess transformer-based.	Strong within-domain results degrade substantially on unseen/cross-domain datasets because of dataset and lexical distribution shifts. All datasets are English, limiting multilingual generalization. Individual models also have specific weaknesses such as RF overfitting, SVM computational cost and KNN weakness in high-dimensional spaces.
10	Tiago A. Almeida; José Maria Gómez Hidalgo; Akebo Yamakami	Contributions to the Study of SMS Spam Filtering: New Collection and Results	2011	SMS Spam Collection: approximately 5,574 English SMS messages labeled ham/spam (about 4,827 ham and 747 spam).	Introduces a public SMS spam corpus and evaluates established content-based classifiers. Multiple tokenization schemes are tested with linear SVM, several Naive Bayes variants, KNN, C4.5, boosted classifiers, PART and MDL. Performance is compared using Spam Caught, Blocked Hams, Accuracy and Matthews Correlation Coefficient; linear SVM provides the strongest baseline.	No dedicated limitation section is stated. The study is based on one relatively small English SMS corpus, so multilingual, temporal concept-drift and cross-corpus generalization are not evaluated.

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PAPERS 11–15

SNO	AUTHORS	TITLE	YEAR	DATASET	METHODOLOGY	LIMITATION
11	Serkan Ballı; Onur Karasoy	Development of Content-Based SMS Classification Application by Using Word2Vec-Based Feature Extraction	2019	SMS Spam Collection dataset.	Builds a Word2Vec model from SMS content and structural message properties. New features measure message distances to spam- and ham-related words. These semantic/structural features are combined with traditional classifiers including Random Forest, MLP, SVM, Logistic Regression and Naive Bayes. Random Forest achieves about 99.64% accuracy.	No explicit limitation section is given. Evaluation is restricted to one English SMS benchmark, and the method is not externally validated on newer or cross-domain spam corpora.
12	Luo Guangjun; Shah Nazir; Habib Ullah Khan; Amin Ul Haq	Spam Detection Approach for Secure Mobile Message Communication Using Machine Learning Algorithms	2020	SMS Spam Collection dataset; 70% training and 30% testing.	Applies Logistic Regression, K-Nearest Neighbors and Decision Tree classifiers to ham/spam SMS classification. The dataset is split into training and testing sets and models are evaluated using accuracy, sensitivity and specificity. Logistic Regression reports the best performance at about 99% accuracy.	No formal limitation section is provided. The experimental scope is limited to a single SMS dataset and three classical classifiers, with no cross-dataset, multilingual or adversarial-spam validation.
13	Pradeep Kumar Roy; Jyoti Prakash Singh; Snehasish Banerjee	Deep Learning to Filter SMS Spam	2020	UCI SMS Spam Collection benchmark: 747 spam and 4,827 non-spam messages.	Compares classical ML classifiers (SGD, Naive Bayes, Logistic Regression, Random Forest and Gradient Boosting) with deep-learning CNN and LSTM models. CNN variants use dropout and 10-fold cross-validation; the best 3-CNN + dropout + 10-fold configuration reaches 99.44% accuracy.	The authors explicitly state that the work is dependent on English-language text messages only; performance on other languages and other spam domains remains for future research.
14	Tian Xia; Xuemin Chen	A Weighted Feature Enhanced Hidden Markov Model for Spam SMS Filtering	2021	UCI SMS Spam Collection plus an additional Chinese SMS dataset for validation.	After tokenization and stop-word removal, each word is weighted according to the difference between its distribution in spam and ham messages. Words are labeled using the sign of the weight and converted into an observation sequence for a discrete Hidden Markov Model. The method is compared with LSTM/CNN and assessed using precision, recall, F1, accuracy and AUC, with emphasis on fast training/filtering.	No dedicated limitation section is stated. The approach still depends on tokenization and word-distribution statistics and is tested mainly on the UCI benchmark plus one Chinese corpus; robustness to temporal concept drift and richer non-text spam is not established.
15	C. Oswald; Sona Elza Simon; Arnab Bhattacharya	SpotSpam: Intention Analysis-Driven SMS Spam Detection Using BERT Embeddings	2022	SMS Spam Collection benchmark dataset.	Uses 13 predefined intention labels to capture message semantics along with textual features. Contextual embeddings are generated using pretrained NLP/BERT-family models, intention scores are computed, and supervised classifiers are used to classify messages as spam or ham. The reported accuracy is about 98.07%, with precision and recall around 0.97.	The authors note that the accuracy is not the best among existing approaches. The method relies on a fixed set of 13 intention labels and a single benchmark dataset, so broader cross-domain and concept-drift behavior remains uncertain.

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PAPERS 16–20

SNO	AUTHORS	TITLE	YEAR	DATASET	METHODOLOGY	LIMITATION
16	Abdallah Ghourab; Manar Alohal	Enhancing Spam Message Classification and Detection Using Transformer-Based Embedding and Ensemble Learning	2023	SMS Spam Collection Dataset.	Cleans SMS text and compares TF-IDF/word embeddings with BERT- and GPT-3-based representations. Four classifiers—SVM, KNN, CNN and LightGBM—are combined using an optimized weighted-voting ensemble. Experiments use an 80/20 train-test split and evaluate accuracy, precision, recall and F1-score; the ensemble with GPT-3 embeddings reaches about 99.91% accuracy.	The study evaluates English SMS messages only. The authors identify multilingual support as future work, so generalization to non-English spam is not demonstrated.
17	Francisco Jáñez-Martino; Rocío Alaiz-Rodríguez; Víctor González-Castro; Eduardo Fidalgo; Enrique Alegre	A Review of Spam Email Detection: Analysis of Spammer Strategies and the Dataset Shift Problem	2023	Experimental study uses Ling-Spam, SpamAssassin, Enron-Spam, TREC07 and CSDMC; Bruce Guenter Spam Archive (2010 and 2018) is used for later/future-data testing.	Combines a literature review of spammer/adversarial strategies with an empirical dataset-shift study. Four text-classification pipelines are formed from BOW or TF-IDF with Naive Bayes or SVM, trained on historical email datasets and tested across different corpora/time periods to quantify generalization degradation.	The empirical study is centered on selected classical text models and mostly historical English email corpora. The paper shows that public datasets from the 2000-2010 era can be unrepresentative of modern spam and that performance can deteriorate severely under dataset shift.
18	Muhammad Adnan; Muhammad Osama Imam; Muhammad Furqan Javed; Iqbal Murtza	Improving Spam Email Classification Accuracy Using Ensemble Techniques: A Stacking Approach	2024	Combined SpamAssassin (6,047 messages) and a 7,582-message subsample of Enron-Spam; merged dataset contains 13,629 emails and is balanced by oversampling.	Preprocesses subject/body text using NLTK and trains Logistic Regression, Decision Tree, Gaussian Naive Bayes, KNN and AdaBoost. Their probability outputs are stacked and passed to a Logistic Regression meta-classifier. Robustness is also checked by changing train/test sizes and base-classifier combinations; stacking achieves about 98.8% accuracy.	The authors identify the need for datasets containing image/personalized spam and emails manipulated by 'email warmer' tools. Continued evolution of spam may reduce effectiveness without new representative training data.
19	Halim Asliyukse; Ozgur Tonkal; Ramazan Kocaoglu	A Comparative Evaluation of a Multimodal Approach for Spam Email Classification Using DistilBERT and Structural Features	2025	Combined corpus of 81,586 emails from seven public spam datasets, including SpamAssassin variants, Enron-Spam, Ling-Spam, two Kaggle spam datasets and a spam-ham dataset used in prior Firefly-optimization research; 31,670 spam and 49,916 ham.	All corpora are normalized to label/text fields and merged, then split stratified 80/20. Class imbalance is handled with random oversampling and class-weighted loss. Classical SVM, Random Forest, Naive Bayes and Logistic Regression are compared with DistilBERT and RoBERTa. A multimodal model fuses DistilBERT semantic representations with structural text features and includes threshold tuning plus temporal robustness analysis.	All datasets are English, restricting multilingual applicability. Adversarial training was not performed because of computational/time constraints, and ambiguous phishing or legitimate marketing language remains challenging.
20	Ye Tian; Xin Dai; Zhijun Li; Hong Guo; Xiao Mao	Improving the Accuracy of Cybersecurity Spam Email Detection Using Ensemble Techniques: A Stacking Approach Machine Learning for Spam Email Detection	2025	Two publicly available sources are integrated: Kaggle email/SMS-style classification dataset with 5,572 samples and the Enron Corpus converted to a compatible category/message structure.	Preprocessing includes cleaning, tokenization, lowercasing, stop-word removal, stemming/lemmatization and engineered length features. TF-IDF is used for textual representation, spam is oversampled to balance classes, and Naive Bayes, KNN, Logistic Regression and XGBoost are combined in a stacking framework with grid-search cross-validation and hyperparameter tuning.	The authors state that the training data are predominantly English and do not represent emerging AI-generated spam. Basic duplication-based oversampling can overfit, and ensemble models add computational cost; future work targets multilingual transfer, transformers and adversarial training.

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DATASET OVERVIEW

PROPERTY	VALUE
Raw shape	5,572 × 5
Useful columns	v1, v2
Target	ham / spam
Duplicates	403
Clean rows	5,169

EDA confirms that the dataset is suitable for a text-classification pipeline and reveals the class imbalance that must be considered during evaluation.

CLEANING PROCESS

- Keep only label and message text columns.
- Rename v1 to label and v2 to message.
- Remove empty unnamed columns and duplicate rows.
- Create length, word, digit, uppercase and punctuation features.
- Prepare the dataset for vectorized classification.

HAM VS SPAM

After cleaning, the dataset contains 4,516 ham messages and 653 spam messages. This shows clear class imbalance, with ham messages being much more frequent than spam messages.

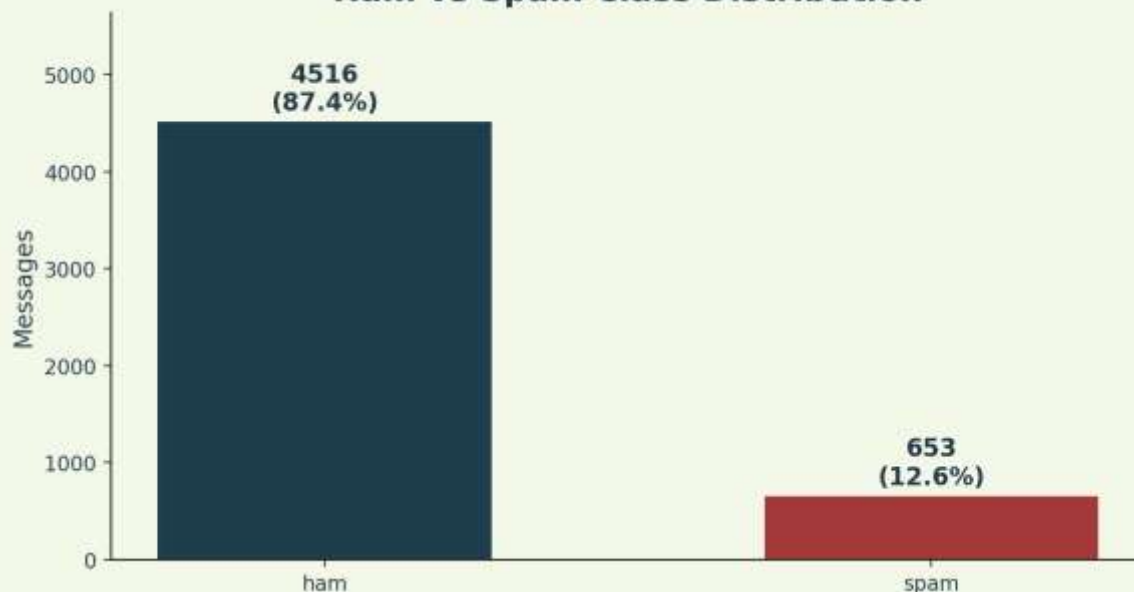
4,516
HAM MESSAGES

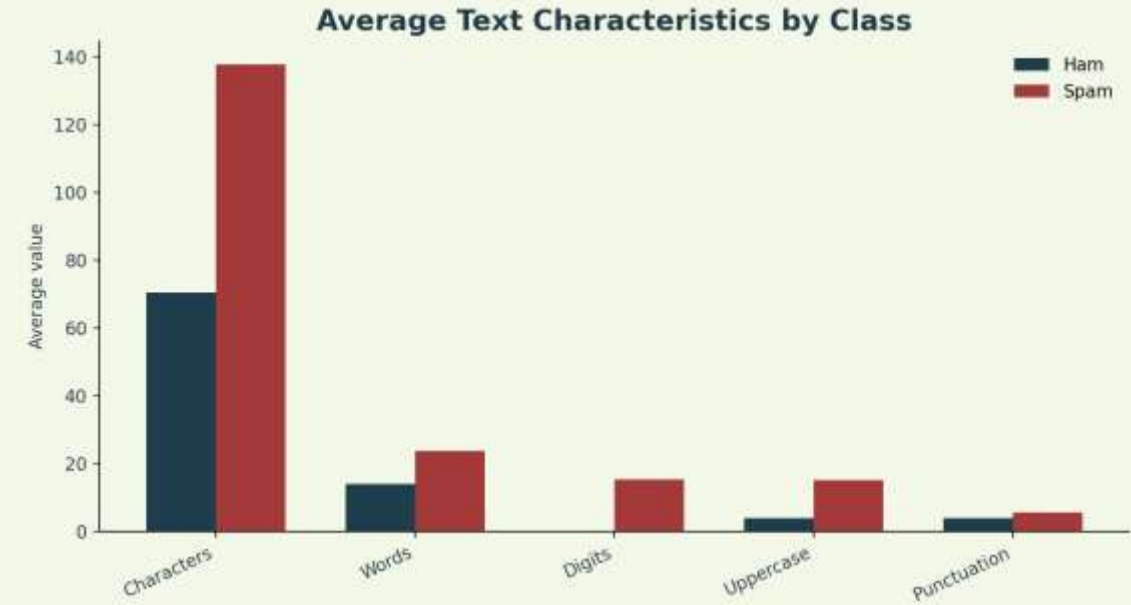
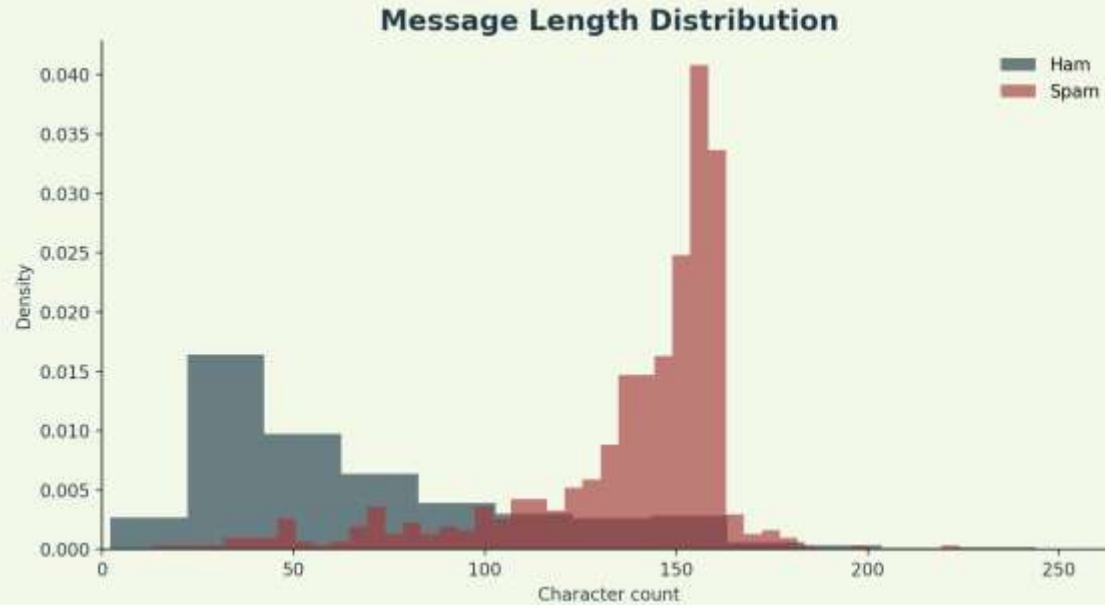
653
SPAM MESSAGES

87:13
APPROX. RATIO

Because of imbalance, Precision, Recall and F1-Score are more meaningful than Accuracy alone.

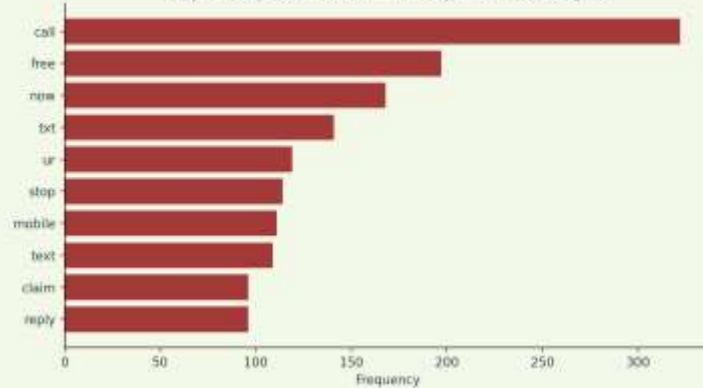
Ham vs Spam Class Distribution



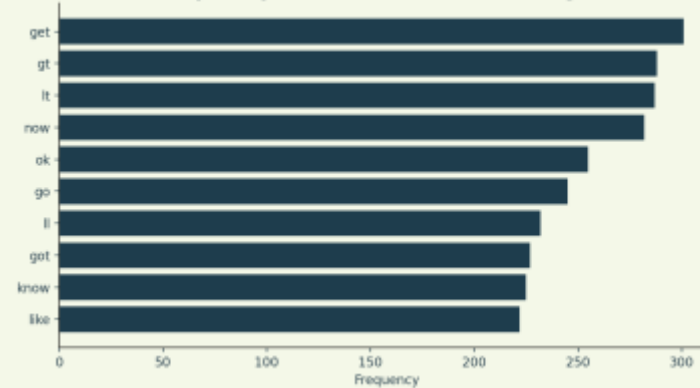


Spam messages generally show stronger text signals such as longer length, more digits, more uppercase characters and more punctuation. These engineered features make EDA meaningful for an NLP dataset.

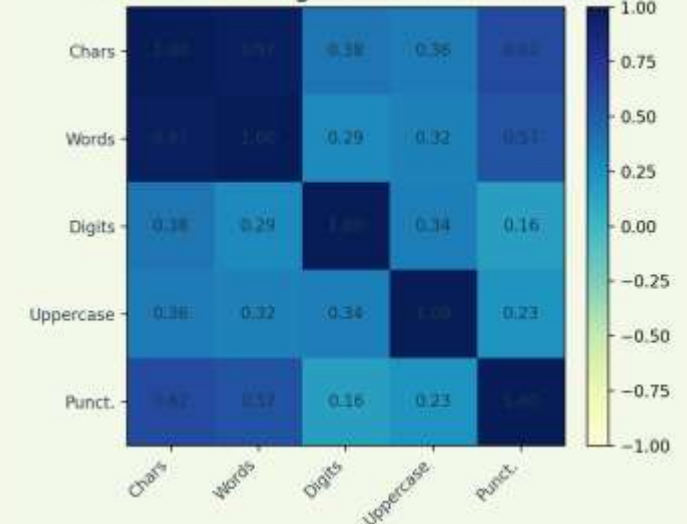
Top Frequent Words in Spam Messages



Top Frequent Words in Ham Messages



Correlation: Engineered Text Features



Frequent-word analysis highlights vocabulary differences between ham and spam. The correlation heatmap is calculated only on engineered numerical text features, making it valid for this text-based dataset.

PROJECT DATASET

- Dataset file: spam.csv
- Raw records: 5,572 labelled messages
- Clean records after duplicate removal: 5,169
- Main columns: label and message
- Labels: ham and spam
- Storage format: CSV

WHY IT FITS

The dataset is suitable for supervised machine learning because every message has a known ham/spam label. Its short text format makes it ideal for preprocessing, vectorization and comparison of Naive Bayes and SVM models. The class imbalance also makes the project academically useful because it requires careful evaluation beyond simple accuracy.

CSV

STORAGE

2

CLASSES

5,169

CLEANED ROWS

NLPPROBLEM
TYPE

CONCLUSION

This project demonstrates how machine learning can be used to classify spam and ham messages from labelled text data. The EDA phase identifies important dataset characteristics such as missing values, duplicates, class imbalance, message length patterns and frequent words. The literature review shows that spam classification has progressed from classical TF-IDF and Naive Bayes models to ensembles, transformers and LLM-based approaches.

Overall, the project provides a clear and practical workflow: clean the text, transform it into numerical vectors, train Naive Bayes and SVM models, and evaluate them using Precision, Recall, F1-Score and Confusion Matrix. The final system can support safer digital communication by detecting unwanted spam messages efficiently.

THANK YOU